Formalization of Weingarten Calculus

Melody Lee Yanting Teng

June 8, 2026

Chapter 1

Schur Weyl Duality

1.1 Operator Space

Suppose $\mathcal{X} = \mathbb{C}^{[d]}$ and $\mathcal{Y} = \mathbb{C}^{[d]}$ where $d \in \mathbb{N}$. Let the set of linear operators be $\mathcal{L}(\mathcal{X}, \mathcal{Y})$. When $\mathcal{Y} = \mathcal{X}$, let the set of linear operators be $\mathcal{L}(\mathcal{X})$.

We briefly restate some relevant definitions and properties of linear operators that we will use in the subsequent sections.

Remark 1.1.1. For any pair of operators $X, Y \in \mathcal{L}(\mathcal{X})$, the Lie bracket $[X, Y] \in \mathcal{L}(\mathcal{X})$ is defined as

$$[X, Y] = XY - YX, \tag{1.1}$$

such that $[X, Y] = 0$ if and only if X and Y commute.

Remark 1.1.2. Let $\mathcal{A} \subseteq \mathcal{L}(\mathcal{X})$ be a subalgebra of $\mathcal{L}(\mathcal{X})$. The commutant of $\mathcal{A}$ is given by

$$\mathrm{Comm}(\mathcal{A}) = \{Y \in \mathcal{L}(\mathcal{X}) : [X, Y] = 0 \text{ for all } X \in \mathcal{A}\}. \tag{1.2}$$

Definition 1.1.3 (Diagonal operators)**.** An operator $X \in \mathcal{L}(\mathcal{X})$ is a diagonal operator if $X_{a,b} = 0$ for all $a, b \in [d]$ where $a \neq b$. For a given vector $u \in \mathcal{X}$, we can define a diagonal operator $\mathrm{Diag}(u) \in \mathcal{L}(\mathcal{X})$ to denote the diagonal operator whose diagonal entries are given by the entries of u. That is, we have

$$\mathrm{Diag}(u)_{a,b} = \begin{cases} u_a & \text{if } a = b \\ 0 & \text{if } a \neq b \end{cases} \tag{1.3}$$

Remark 1.1.4 (Projection operators). Let Π denote the project operator onto a subspace $\mathcal{V}$. Then Π is a linear operator that satisfies the following properties:

- $\Pi^2 = \Pi$, the idempotent property; and
- $\Pi^* = \Pi$, the self-adjoint property.

Let the collection of all projection operators be denoted $\mathrm{Proj}(\mathcal{X})$. There is a uniquely defined projection operator $\Pi \in \mathrm{Proj}(\mathcal{X})$ such that its image $\mathrm{Im}(\Pi) = \mathcal{V}$. When necessary, we denote this projection operator as $\Pi_{\mathcal{V}}$ to indicate the subspace $\mathcal{V}$ onto which it projects.

Remark 1.1.5 (Isometries). An operator $\mathcal{A} \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ is an isometry if it satisfies the condition

$$\mathcal{A}^* \mathcal{A} = I_{\mathcal{X}}, \tag{1.4}$$

which is equivalent to preserving the Euclidean norm of vectors in $\mathcal{X}$, i.e., $\|\mathcal{A}|\psi\rangle\| = \||\psi\rangle\|$ for all $|\psi\rangle \in \mathcal{X}$.

Remark 1.1.6 (Unitary operators). Denote the set of isometries that map $\mathcal{X}$ to itself as the set of unitary operators $U(\mathcal{X})$. Every unitary operator $U \in U(\mathcal{X})$ is necessarily invertible and satisfies the normal condition

$$UU^* = U^*U = I_{\mathcal{X}}. \tag{1.5}$$

1.2 Symmetric subspaces

Definition 1.2.1 (Symmetric subspace of linear operators)**.** The symmetric subspace of linear operators $\mathcal{L}(\mathcal{X})^{\otimes k}$ is the set of all operators $X \in \mathcal{L}(\mathcal{X}^{\otimes k})$ that remain unchanged (invariant) under the permutation W_π. That is,

$$\mathcal{L}(\mathcal{X})^{\otimes k} = \{X \in \mathcal{L}(\mathcal{X}^{\otimes k}) : X = W_\pi X W_\pi^* \text{ for all } \pi \in S_k\}. \tag{1.6}$$

Remark 1.2.2. Let S_k denote the symmetric group. Then given $\pi \in S_k$, the permutation matrix (operator) $V_d(\pi)$ is the unitary matrix that satisfies

$$W_\pi |\psi_1\rangle \otimes \dots |\psi_k = |\psi_{\pi^{-1}(1)}\rangle \otimes \cdots \otimes |\psi_{\pi^{-1}(k)}\rangle \tag{1.7}$$

for all $|\psi_1\rangle, \dots, |\psi_k\rangle \in \mathbb{C}^d$.

Definition 1.2.3 (Standard computational basis of linear operators)**.** The standard computational basis of linear operators $\{E_{i,j}\}_{i,j\in[d]}$ is defined as the set of d^2 operators in $\mathcal{L}(\mathcal{X})$ such that

$$E_{i,j} = |i\rangle\langle j| \tag{1.8}$$

for all $i, j \in [d]$, where $\{|i\rangle\}_{i\in[d]}$ is the standard computational basis of $\mathcal{X}$.

Definition 1.2.4 (Operator-vector correspondence)**.** The operator-vector correspondence is a bijective mapping between the set of linear operators $\mathcal{L}(\mathcal{X})$ and the tensor product space $\mathcal{X} \otimes \mathcal{X}$ defined as

$$\text{vec} : \mathcal{L}(\mathcal{X}) \to \mathcal{X} \otimes \mathcal{X}, \quad E_{i,j} \mapsto |i\rangle \otimes |j\rangle \text{ for all } i, j \in [d]. \tag{1.9}$$

One relevant identity we use is that for $A_0, A_1, B \in \mathcal{L}(\mathcal{X})$,

$$(A_0 \otimes A_1)\text{vec}(B) = \text{vec}(A_0 B A_1^{\mathrm{T}}). \tag{1.10}$$

1.3 Permutation Invariance and Properties of Unitarily Invariant Measures

Theorem 1.3.1 (Tensor power span theorem)**.** *Let k be a positive integer, and let $X \in \mathbb{C}^[d]$. For any set $\mathcal{A} \subseteq \mathbb{C}$ satisfying $|\mathcal{A}| \geq k+1$, it holds that*

$$Span\{v^{\otimes k} : v \in \mathcal{A}^{[d]}\} = \mathcal{X}^{\otimes k}. \tag{1.11}$$

Proposition 1.3.2 (Symmetric subspace and span of tensor product space)**.** *Let $\mathcal{X}$ be a complex Euclidean space and k be a positive integer. Then the following statements are equivalent:*

1. $X \in \mathcal{L}(\mathcal{X})^{\otimes k}$.

2. *For $V \in U(\mathcal{X}^{\otimes k} \otimes \mathcal{X}^{\otimes k}, (\mathcal{X} \otimes \mathcal{X})^{\otimes k})$ being the isometry defined by*

$$V\, vec(Y_1 \otimes \cdots \otimes Y_k) = vec(Y_1) \otimes \ldots vec(Y_n) \tag{1.12}$$

   holding for all $Y_1, \ldots, Y_k \in \mathcal{L}(\mathcal{X})$, one has that

$$V\, vec(X) \in (\mathcal{X} \otimes \mathcal{X})^{\circledS k}. \tag{1.13}$$

3. *$X \in Span\{Y^{\otimes k} : Y \in \mathcal{L}(\mathcal{X})\}$*

Theorem 1.3.3 (Equivalence of symmetric subspace and unitary tensor span)**.** *Let $\mathcal{X}$ be a complex Euclidean space and k be a positive integer. Then the symmetric subspace of linear operators is equivalent to the span of the unitary tensor products, i.e.,*

$$\mathcal{L}(\mathcal{X})^{\circledS k} = Span\{U^{\otimes k} : U \in \mathcal{U}(\mathcal{X})\}. \tag{1.14}$$

Lemma 1.3.4 (Invariant subspace lemma)**.** *Let $\mathcal{X}$ be a complex Euclidean space, let $\mathcal{V} \subseteq \mathcal{X}$ be a subspace, and let $A \in \mathcal{L}(\mathcal{X})$ be an operator. The following statements are equivalent:*

1. *It holds that both $A\mathcal{V} \subseteq \mathcal{V}$ and $A^*\mathcal{V} \subseteq \mathcal{V}$.*
2. *It holds that $[A, \Pi_{\mathcal{V}}] = 0$.*

Theorem 1.3.5 (Double Commutant Theorem)**.** *Let $\mathcal{A}$ be a self-adjoint, unital subalgebra of $\mathcal{L}(\mathcal{X})$. Then the double commutant of $\mathcal{A}$ is equal to $\mathcal{A}$, i.e.,*

$$Comm(Comm(\mathcal{A})) = \mathcal{A}. \tag{1.15}$$

Proof. For all $X \in \text{Comm}(\mathcal{A})$ and any $A \in \mathcal{A}$, we have

$$[X, A] = AX - XA = 0. \tag{1.16}$$

Therefore, $A \in \text{Comm}(\text{Comm}(\mathcal{A}))$, so $\mathcal{A} \subseteq \text{Comm}(\text{Comm}(\mathcal{A}))$.

To demonstrate reverse inclusion, consider the subalgebra $\mathcal{B} \subseteq \mathcal{L}(\mathcal{X} \otimes \mathcal{X})$ defined as

$$\mathcal{B} = \{X \otimes \mathbb{I} : X \in \mathcal{A}\}. \tag{1.17}$$

For all $Y \in \mathcal{L}(\mathcal{X} \otimes \mathcal{X})$, it can be written in terms of the standard computational basis

$$Y = \sum_{i,j \in [d]} \sum_{a,b \in [d]} c_{i,j,a,b} |i\rangle\langle j| \otimes |a\rangle\langle b|. \tag{1.18}$$

Using the definition of the standard computational basis of operators, we can rearrange the right-hand side into the form

$$Y = \sum_{a,b \in [d]} Y_{a,b} \otimes E_{a,b} \tag{1.19}$$

where $Y_{a,b} = \sum_{i,j \in [d]} c_{i,j,a,b} |i\rangle\langle j|$.

Y commutes with $\mathcal{B}$ if and only if $Y(X \otimes \mathbb{I}) = (X \otimes \mathbb{I})Y$ for all $X \in \mathcal{A}$. Substituting Equation 1.19 and simplifying shows that Y commutes with $\mathcal{B}$ if and only if $[Y_{a,b}, X] = 0$. For any $X \in \text{Comm}(\text{Comm}(\mathcal{A}))$, X must commute with $\text{Comm}(\mathcal{A})$, and therefore

$$X \otimes \mathbb{I} \in \text{Comm}(\text{Comm}(\mathcal{B})). \tag{1.20}$$

It remains to show that every element in $\mathrm{Comm}(\mathrm{Comm}(\mathcal{B}))$ is of the form $X \otimes \mathbb{I}$. Define a subspace $\mathcal{V} \subseteq X \otimes X$, where

$$\mathcal{V} = \left\{ \mathrm{vec}(X) = \sum_{a,b \in [d]} c_{a,b} |a\rangle \otimes |b\rangle : X \in \mathcal{A} \right\}. \tag{1.21}$$

For any arbitrary $X \in \mathcal{A}$, we find that

$$\begin{aligned}
(X \otimes \mathbb{I})\mathcal{V} &= \{(X \otimes \mathbb{I})V : V \in \mathcal{V}\} \\
&= \left\{ (X \otimes \mathbb{I}) \left(\sum_{a,b \in [d]} c_{a,b} |a\rangle \otimes |b\rangle \right) : \sum_{a,b \in [d]} c_{a,b} |a\rangle\langle b| \in \mathcal{A} \right\} \\
&= \left\{ \sum_{a,b \in [d]} c_{a,b} X |a\rangle \otimes |b\rangle : \sum_{a,b \in [d]} c_{a,b} |a\rangle\langle b| \in \mathcal{A} \right\} \\
&\subseteq \mathcal{V},
\end{aligned} \tag{1.22}$$

the inclusion of which follows from the fact that $\mathcal{A}$ is an algebra so $X|a\rangle = Xe_a \in \mathcal{A}$. Similarly, since $\mathcal{A}$ is self-adjoint,

$$(X^* \otimes \mathbb{I})\mathcal{V} \subseteq \mathcal{V}. \tag{1.23}$$

By Lemma 1.3.4, this implies $[X \otimes \mathbb{I}, \Pi_{\mathcal{V}}] = 0$, and hence

$$\Pi_{\mathcal{V}} \in \mathrm{Comm}(\mathcal{B}). \tag{1.24}$$

Therefore, for any arbitrary $X \in \mathrm{Comm}(\mathrm{Comm}(\mathcal{A}))$, Equation 1.20 implies

$$X \otimes \mathbb{I} \in \mathrm{Comm}(\mathrm{Comm}(\mathcal{B})). \tag{1.25}$$

It follows from Equation 1.24 that

$$[X \otimes \mathbb{I}, \Pi_{\mathcal{V}}] = 0. \tag{1.26}$$

By Lemma 1.3.4, this yields

$$(X \otimes \mathbb{I})\mathcal{V} \subseteq \mathcal{V}. \tag{1.27}$$

Since the subalgebra $\mathcal{A}$ is unital, $\mathbb{I} \in \mathcal{A}$ and thus $\mathrm{vec}(\mathbb{I}) \in \mathcal{V}$. Therefore,

$$\begin{aligned}
(X \otimes \mathbb{I})\mathrm{vec}(\mathbb{I}) &= (X \otimes \mathbb{I}) \sum_{a \in [d]} |a\rangle \otimes |a\rangle \\
= \sum_{a \in [d]} X|a\rangle \otimes |a\rangle &= \sum_{a \in [d]} \left(\sum_{b,c \in [d]} X_{b,c} |b\rangle\langle c| \right) |a\rangle \otimes |a\rangle \\
&= \sum_{a \in [d]} \sum_{b \in [d]} X_{b,a} |b\rangle \otimes |a\rangle \\
&= \sum_{b,a \in [d]} X_{b,a} |b\rangle \otimes |a\rangle = \mathrm{vec}(X) \in \mathcal{V}.
\end{aligned} \tag{1.28}$$

By the bijectivity of vec, $X \in \mathcal{A}$, so $\mathrm{Comm}(\mathrm{Comm}(\mathcal{A})) \subseteq \mathcal{A}$. ◻

Theorem 1.3.6 (Schur-Weyl duality)**.** *Let $\mathcal{X}$ be a complex Euclidean space, k be a positive integer, and $X \in \mathcal{L}(\mathcal{X}^{\otimes k})$ be an operator. Then the set of all commutators of the unitary group is equivalent to the span of the permutation operators, i.e.,*

$$\mathit{Comm}(\mathcal{U}(\mathcal{X}), k) = \mathit{Span}\{W_\pi : \pi \in S_k\} \tag{1.29}$$

Proof. Let $X \in \mathrm{Comm}(\mathcal{U}(\mathcal{X}), k)$. then by the bilinearity of the Lie bracket,

$$X \in \mathrm{Comm}(\mathrm{Span}\{U^{\otimes k} : U \in U(\mathcal{X})\}) \tag{1.30}$$

Proposition 1.3.2 implies that $X \in \mathrm{Comm}(\mathcal{L}(\mathcal{X})^{\circledS k})$, so

$$\mathrm{Comm}(\mathrm{Span}\{U^{\otimes k} : U \in U(\mathcal{X})\}) = \mathrm{Comm}(\mathcal{L}(\mathcal{X})^{\circledS k}). \tag{1.31}$$

By Definition 1.2.1, $Z \in \mathcal{L}(\mathcal{X})^{\circledS}$ if and only if there exists some $X \in \mathcal{L}(\mathcal{X})$ such that

$$Z = X = W_\pi X W_\pi^* \text{ for all } \pi \in S_k. \tag{1.32}$$

Right-multiplying and rearranging by W_π yields $XW_\pi - W_\pi X = 0$, and so by the bilinearity of the Lie bracket,

$$Z = X \in \mathrm{Comm}(\mathrm{Span}\{W_\pi : \pi \in S_k\}). \tag{1.33}$$

Hence

$$\mathcal{L}(\mathcal{X})^{\circledS k} = \mathrm{Comm}(\mathrm{Span}\{W_\pi : \pi \in S_k\}). \tag{1.34}$$

By Theorem 1.3.5, taking the commutant of both sides yields

$$\mathrm{Comm}(\mathcal{L}(\mathcal{X})^{\circledS k}) = \mathrm{Span}\{W_\pi : \pi \in S_k\}. \tag{1.35}$$

By substitution of Eq. 1.31, we have shown

$$\mathrm{Comm}(U(\mathcal{X}), k) = \mathrm{Span}\{W_\pi : \pi \in S_k\}. \tag{1.36}$$

□

1.4 Examples

We illustrate some simple examples of Haar measure computations. In particular, we are interested in first-order computations on discrete and continuous groups. We approach this from the perspective of group averaging and twirling.

Definition 1.4.1 (Group average)**.** Let G be a group of order N and R a representation of the group. Define the group average to be

$$\langle X \rangle_G = \frac{1}{N} \sum_g R(g) X R(g)^\dagger. \tag{1.37}$$

Remark 1.4.2 (Assumptions)*.* Consider a n-qubit system. Let $d = 2^n$, and suppose we seek to compute moments of order k. We assume for now that $d > k$ for the examples below.

1.4.1 Pauli twirling

The end goal of this section is to prove the Irrep Group Averaging Corollary

$$\langle X \rangle_G = \frac{1}{d} \mathrm{Tr}[X] I, \tag{1.38}$$

where $n_a = d$ is the dimension of the vector space of the representation. To illustrate this, we first consider the group average of the single-qubit Pauli group on some arbitrary initial state ρ.

Remark 1.4.3 (Pauli groups). We denote the single-qubit Pauli group as

$$G = \{\pm(i)\sigma_x, \pm(i)\sigma_y, \pm(i)\sigma_z, \pm(i)I\}, \tag{1.39}$$

where the Pauli matrices are given by

$$\sigma_0 = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \sigma_x = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \sigma_y = Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \sigma_z = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \tag{1.40}$$

We note the following useful expressions.

Remark 1.4.4 (Bloch vectors). For any single-qubit quantum state ρ, we can write it in terms of its Bloch vector,

$$\rho = \frac{1}{2}(I + r \cdot \sigma), \tag{1.41}$$

where $r = (r_x, r_y, r_z) \in \mathbb{R}^3$ is the coordinate of the quantum state on the Bloch sphere and $\sigma = (\sigma_x, \sigma_y, \sigma_z)$ is a three-tuple of the Pauli operators.

Remark 1.4.5 (Pauli group identities). For all non-identity σ_i, σ_j in the Pauli group where $i \neq j$, the following three statements are true.

1. $\sigma_i \sigma_j \sigma_i = -\sigma_j$,
2. $\sigma_j^3 = \sigma_j$, and
3. $\sigma_i^2 = \sigma_i$, since σ_i is represented by a self-adjoint matrix.

We want to show the following simpler equivalence using matrix computations.

Theorem 1.4.6 (Pauli twirling of a single qubit)**.** *For an arbitrary single-qubit state ρ,*

$$\langle\rho\rangle_G = \frac{1}{N}\sum_g R(g) X R(g)^\dagger = \frac{I}{2}, \tag{1.42}$$

the maximally mixed state (since it is proportional to the identity matrix).

Proof. Note that G has order $N = 4$. For any Pauli operator $\pm(i)X \in G$, X is self-adjoint. Thus, we have that

$$\pm(i)X\rho(\pm(i)X)^\dagger = (\pm i)(\mp i)X\rho X = X\rho X. \tag{1.43}$$

By Def. 1.4.1 and Eq. 1.43,

$$\langle\rho\rangle_G = \frac{1}{4}\left(\sigma_x\rho\sigma_x + \sigma_y\rho\sigma_y + \sigma_z\rho\sigma_z + I\rho I\right). \tag{1.44}$$

By Remark 1.4.4 and Remark 1.4.5, we can rewrite ρ such that

$$\begin{aligned}
\langle\rho\rangle_G &= \frac{1}{4}\left(\sigma_x(\frac{1}{2}(I + r\cdot\sigma)\sigma_x + \sigma_y(\frac{1}{2}(I + r\cdot\sigma))\sigma_y + \sigma_z(\frac{1}{2}(I + r\cdot\sigma))\sigma_z + I(\frac{1}{2}(I + r\cdot\sigma))I\right) \\
&= \frac{1}{4}\left(\frac{1}{2}\left[\sigma_x I\sigma_x + r_x\sigma_x^3 + r_y\sigma_x\sigma_y\sigma_x + r_z\sigma_x\sigma_z\sigma_x\right] + \frac{1}{2}\left[\sigma_y I\sigma_y + r_x\sigma_y\sigma_x\sigma_y + r_y\sigma_y^3 + r_z\sigma_y\sigma_z\sigma_y\right]\right. \\
&\quad \left. + \frac{1}{2}\left[\sigma_z I\sigma_z + r_x\sigma_z\sigma_x\sigma_z + r_y\sigma_z\sigma_y\sigma_z + r_z\sigma_z^3\right] + \frac{1}{2}\left[I^3 + r_x I\sigma_x I + r_y I\sigma_y I + r_z I\sigma_z I\right]\ \right) \\
&= \frac{1}{8}\big(\ [I + r_x\sigma_x - r_y\sigma_y - r_z\sigma_z] + [I - r_x\sigma_x + r_y\sigma_y - r_z\sigma_z] \\
&\quad + [I - r_x\sigma_x - r_y\sigma_y + r_z\sigma_z] + [I + r_x\sigma_x + r_y\sigma_y + r_z\sigma_z]\ \big) \\
&= \frac{1}{8}(4I) = \frac{I}{2}.
\end{aligned} \tag{1.45}$$

□